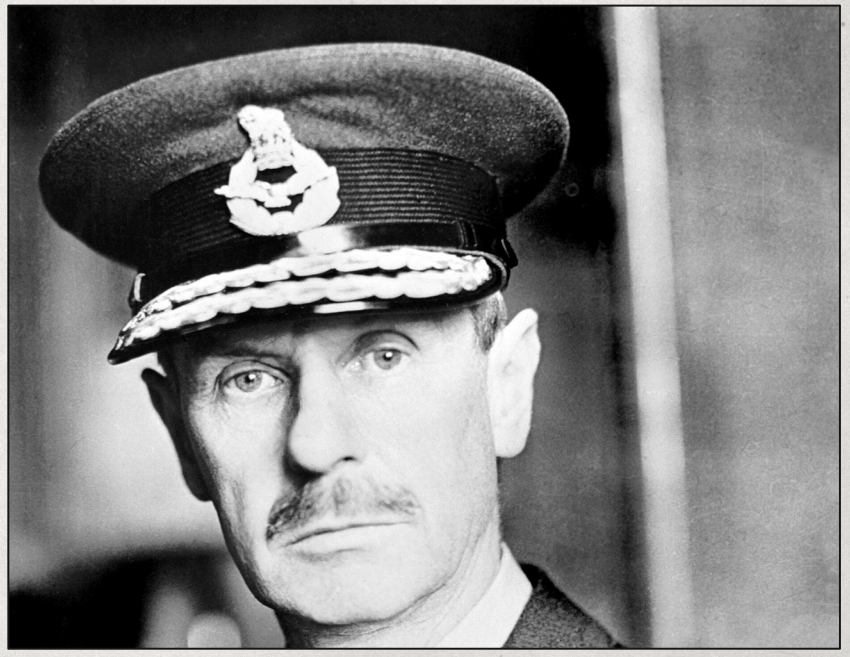
Tomorrow's Leaders

Special edition: effective communication — Hugh Dowding and the Battle of Britain, 1940

The battle won by information

How Hugh Dowding's air defence network got each raid report to the right person in time, and what a student team can learn from it

By Group [number]: [member names]



Air Chief Marshal Sir Hugh Dowding, Commander-in-Chief of RAF Fighter Command
Imperial War Museums, D 1417 (public domain)

Britain stands alone

Summer 1940: France has fallen, and the German air force is just across the Channel



Spitfires of No. 610 Squadron from Biggin Hill in "vic" formation, 24 July 1940
Imperial War Museums, CH 740 (public domain)

10 May Germany attacks France, Belgium and the Netherlands.

22 June France signs an armistice with Germany.

10 July The Luftwaffe begins its attack on Britain. Germany cannot invade until it controls the air.

If you commanded Britain's fighters, where would you send them?

Tomorrow's Leaders

Part 1

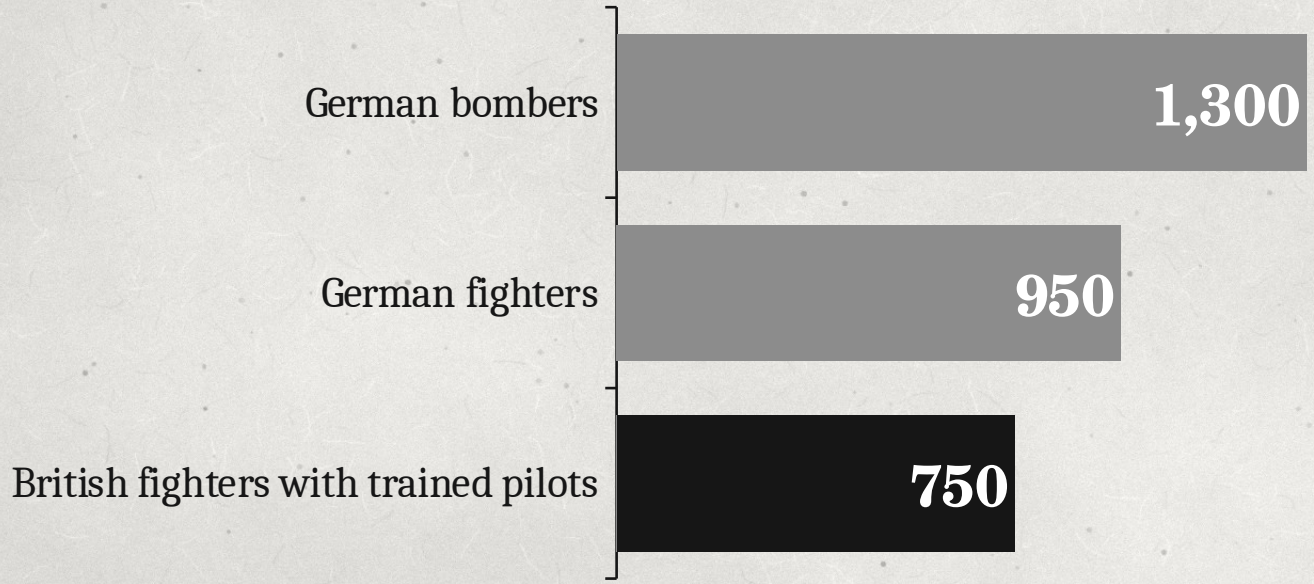
The story

How did Britain meet every raid in time, with no fighters to spare?

1

750 British fighters, 2,250 German aircraft

Strength on 10 August 1940, counting only aircraft ready to fly



German figures are "over" these numbers. Britain had 1,100 serviceable fighters, but trained pilots for only about 750.

Fighter against fighter, the two sides were close. But Britain's fighters also had to find and stop more than 1,300 bombers.

A squadron sent to the wrong place could not reach the real raid in time.

Gleave (1991, p. 48)

Twenty minutes to meet each raid

11 Group warned its controllers that raiders could be over London 20 minutes after the first radar plot

20 min

minutes of warning from a Kent radar station for a raid forming over France

4 min

minutes for the network to mark that raid on the operations room table

2 min

minutes, at times, for the 11 Group controller to decide

Every minute spent passing a report along was a minute less for the fighters to reach the raid.

"The Battle of Britain" (2015); Battle of Britain Bunker (n.d.)

Seeing the raid was not enough

A radar operator sees 100 aircraft crossing the Channel. What happens next?

Option A

Phone the nearest airfield

A few fighters take off alone. Nobody else knows where the raid is going, and other airfields wait for orders.

Option B

Report to headquarters

Each level studies the report and passes an order down. The order reaches the pilots after the bombs have fallen.

The hard part was not seeing the raid. It was getting the right report to the right person in time.

Thought experiment based on the timings on slide 5.

The man who built the network

Air Chief Marshal Hugh Dowding, head of RAF Fighter Command from 1936 to 1940



Dowding (centre, hands behind his back) as King George VI decorates a pilot at Hornchurch, 27 June 1940
Imperial War Museums, CH 442 (public domain)

1935

As the RAF's head of research, he backs radar after a successful test near Daventry.

1936

He becomes the first head of Fighter Command, at Bentley Priory near London.

1936 to 1940

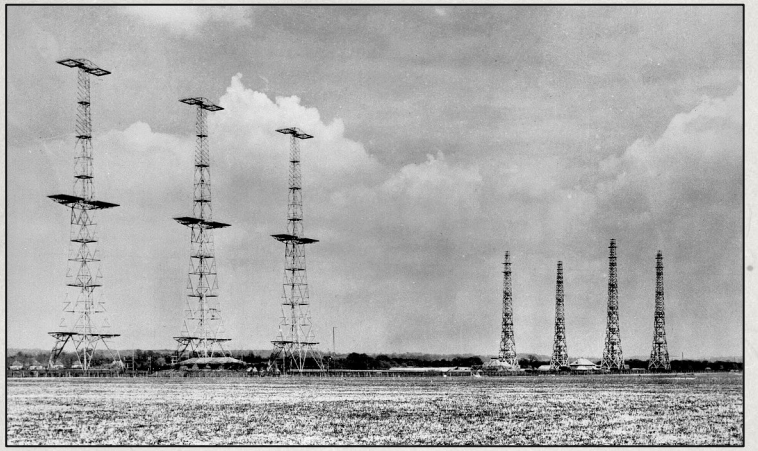
He links radar stations, observers, operations rooms and pilots into one network.

His idea, in our words: send each squadron to a raid that someone has already reported, instead of sending it to search the sky.

Bournemouth University (n.d.); Royal Air Force Museum (n.d.-a); Imperial War Museums (n.d.-c)

Step 1: Report

Radar stations faced out to sea, so volunteers watched the land



Chain Home radar station at Poling, Sussex, photographed in 1945
Imperial War Museums, CH 15173 (public domain)



An observer reports by chest telephone at Kings Langley, 1941 to 1945
Imperial War Museums, CH 8215 (public domain)

Chain Home radar

About 20 stations on the coast. They could not track aircraft over land, and their plots were not always accurate.

Observer Corps

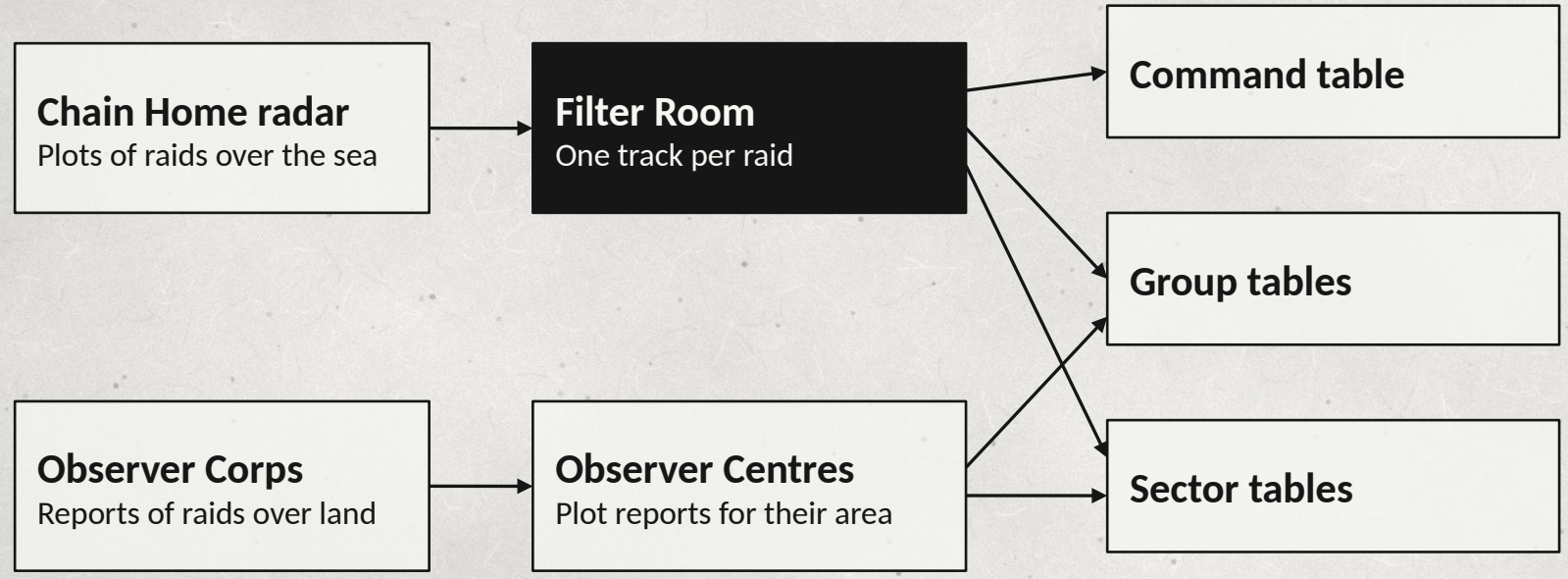
About 30,000 volunteers at about 1,000 posts. They spotted aircraft by eye and reported them by telephone.

Next problem Several radar plots often described the same raid, and some disagreed about its position or strength.

Historic England (2020); Royal Air Force Museum (n.d.-b); Imperial War Museums (n.d.-b)

Step 2: Filter

In the Filter Room, staff compared the radar plots and kept one track per raid



At the same time

Each filtered track went by telephone to the Command, Group and Sector tables at the same moment, so every level saw the same raid.

Next problem Commanders at three levels had to read the same table quickly and know how old each plot was.

Dowding (1946); Imperial War Museums (n.d.-c)

Step 3: Display

Every operations table used the same symbols



The operations room at Fighter Command headquarters, Bentley Priory (date not recorded)
Imperial War Museums, C 1869 (public domain)



Sector clock

Its red, yellow and blue segments each span 5 minutes. Plotters used markers of the same colours, so a controller could see how old each plot was.



Raid block

A block on the table showed the raid's name and estimated strength, and an arrow showed its direction.



Tote board

A board showed the state of each squadron, such as “at readiness” or “left ground”.

Next problem A pilot in a cockpit needed his orders in a few words.

Royal Air Force Museum (n.d.-c); Imperial War Museums (2020, n.d.-c)

Step 4: Order

A sector controller gave each squadron its course and height by radio, in a few code words



Pilots of No. 19 Squadron stage a "scramble" for the camera at Fowlmere, September 1940
Imperial War Museums, CH 1398 (public domain)

Code word	Meaning
Scramble	Take off
Vector 230	Fly a course of 230 degrees
Angels 20	Fly at 20,000 feet
Pancake	Land

Group decided which sector met each raid and with how many fighters. The sector controller directed the pilots.

Dowding (1946)

"We had nothing like it"

Adolf Galland, a German fighter commander in 1940, on the British network

---

"From the first the British had an extraordinary advantage, never to be balanced out at any time during the whole war, which was their radar and fighter control network and organisation. It was for us a very bitter surprise. We had nothing like it."

Galland (as cited in Wood, 1991)

The Luftwaffe bombs the network

From late August, German bombers attacked the airfields of the sector stations

At the end of August and in early September, the Luftwaffe bombed RAF airfields in south-east England, including the sector stations whose controllers directed the fighters. After one raid, Kenley's operations room had to move to a butcher's shop in Caterham.

On 7 September, Hitler moved the main attack to London. One reason was a false German estimate that Fighter Command had only 150 to 300 fighters left.

What the switch meant

London suffered heavy bombing, while Fighter Command repaired its airfields. Boog calls the switch a great relief for Fighter Command, though less decisive than often believed.

Boog (1991)

“There are none”

15 September 1940: Churchill watches the plotting table at No. 11 Group, Uxbridge

“I now asked, ‘What other reserves have we?’
‘There are none,’ said Air Vice Marshal Park.”

Winston Churchill, recalling the visit. Park commanded No. 11 Group.

About 56 to 60 German aircraft were lost that day. On 17 September, Hitler postponed the invasion of Britain until further notice.



The preserved No. 11 Group operations room at Uxbridge, photographed in 2010
Photo: Danielstirland, CC0, via Wikimedia Commons

Churchill (as cited in Glines, 1990); Price (2003)

What the story shows

Two scholars' definitions fit the network, but neither mentions time

Keyton (2011)

Communication is “transmitting information and common understanding from one person to another”.

In the network
One track per raid reached every table, and every reader understood the same symbols.

Spitzberg (2013)

Competence is an impression of effectiveness and appropriateness, formed by the people you communicate with.

In the network
“Vector 230, Angels 20” told a pilot where to fly, in a code short enough for a radio in combat.

Our definition

Effective communication gets the right information to the right person in time, in a form that person can act on.

Keyton (2011, as cited in Lunenburg, 2010); Spitzberg (2013)

Tomorrow's Leaders

Part 2

Two theories

Why did the network work?

2

Theory 1: The 7 Cs

Murphy, Hildebrandt and Thomas (1997) list seven qualities of a good message; we phrase each as a question

Clarity

What is the one main point?

Correctness

Are the facts and tone right?

Conciseness

Can it be said in fewer words?

Concreteness

Is it backed by specific facts?

Completeness

Does the reader have all they need to act?

Consideration

Is it written from the reader's side?

Courtesy

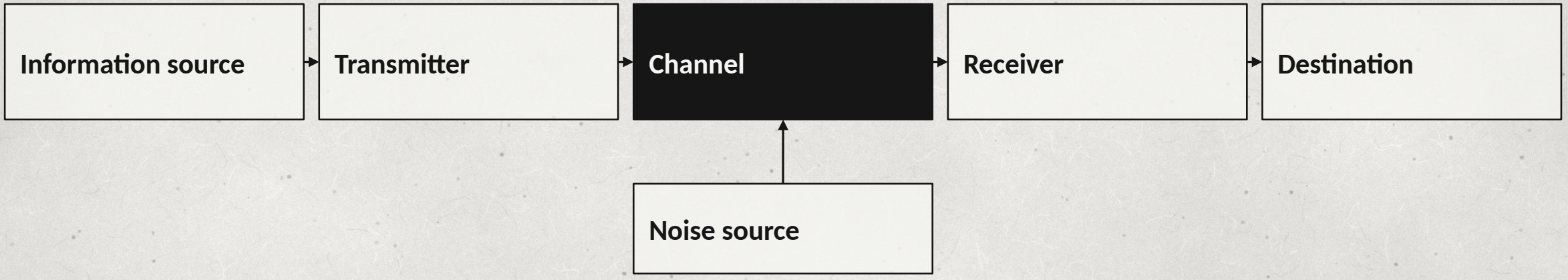
Is it respectful, even when firm?

Best suited to one message from one sender

Murphy et al. (1997)

Theory 2: Shannon and Weaver

A message travels from a source to a destination, and noise can distort it on the way



Weaver's three levels of communication problems

A. Technical
Can the symbols be sent accurately?

B. Semantic
Do the symbols carry the intended meaning?

C. Effectiveness
Does the meaning change behaviour as intended?

Shannon & Weaver (1949)

Comparing the two theories

The 7 Cs apply to one message, while Shannon and Weaver's model follows a message through a network

	7 Cs	Shannon and Weaver
Focus	The quality of one message	The path of a message, and where noise enters
Strength	A quick checklist anyone can apply	Shows where a message can be lost or distorted
Weakness	A checklist rather than a tested theory	One-way: no feedback, and nothing on the relationship between people
Culture	Assumes meaning is in the explicit words, which suits low-context cultures	Also assumes meaning travels in the message, not in the context

Murphy et al. (1997); Shannon & Weaver (1949); Craig (1999); Hall (1976); Cardon (2008)

Tomorrow's Leaders

Part 3

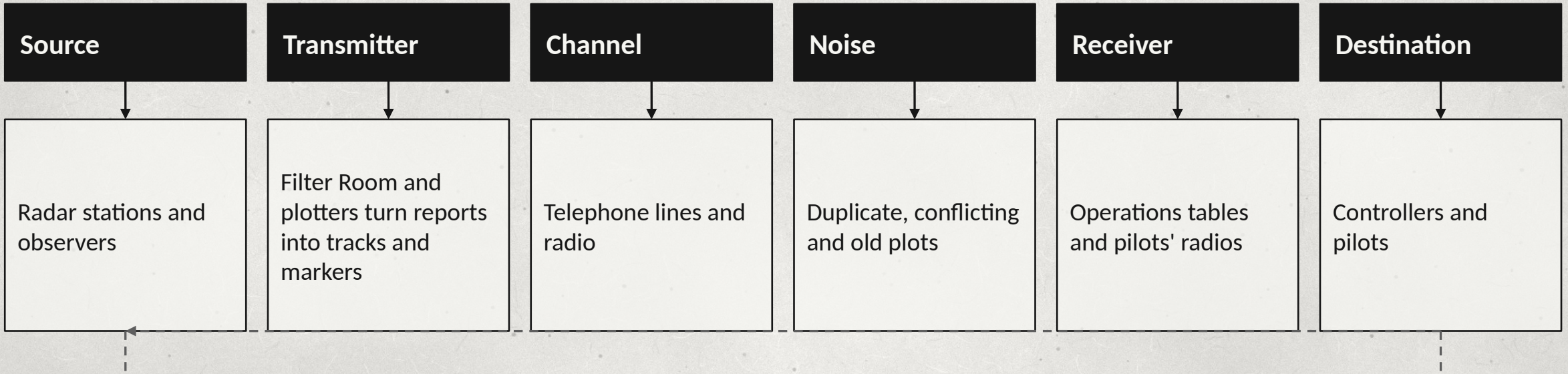
The theories applied

How do the theories explain the network, and what do they miss?

3

The network, part by part

Each part of the model matches a part of the network, and the network added the feedback the model leaves out



Feedback, not in the original model: radio position fixes and pilots' reports

The network reduced all three of Weaver's problems but did not remove them: radar plots were not always accurate, and commanders still disagreed.

Shannon & Weaver (1949); Dowding (1946); Imperial War Museums (n.d.-c)

One radio order against the 7 Cs

An example order built from the code words in Dowding's despatch: "Scramble. Vector 230. Angels 20."

Clarity	●●●	One action: take off and fly to the raid
Correctness	●●●	Course and height come from the filtered track
Conciseness	●●●	Five words
Concreteness	●●●	An exact course and an exact height
Completeness	●●●	When, where and how high to fly
Consideration	●●●	Only words a pilot can act on while flying
Courtesy	●○○	***No "please": in combat, speed came first***

19 of 21

By the checklist, the order fails only on courtesy: it has no "please". In a cockpit, a short order respects the pilot's time, so courtesy depends on a situation the checklist does not ask about.

Dowding (1946); Murphy et al. (1997). Scores are our group's judgment.

Three limits of our analysis

The theories leave out disagreement, our story credits one man, and most of our sources are British

1 Same network, different tactics

Keith Park and Trafford Leigh-Mallory, two of Dowding's group commanders, received reports from the same network and still disagreed over how to use their squadrons.

2 Many people built the network

It built on First World War air defence and scientists' radar work, and thousands of plotters and observers ran it.

3 Our sources lean one way

Most of our sources are British or RAF institutions, and Galland gave his verdict after the war.

If thousands of people ran the network, how much of its success belongs to Dowding?

Imperial War Museums (n.d.-a, n.d.-c); Wood (1991)

Tomorrow's Leaders

Part 4

Lessons for us

How can a student team get the right information to the right person in time?

4

Our group chats have the same problem

Amy posts in a five-person group chat: "Can someone do the theory part?" What happens next?

Option A

Nobody replies

Each member thinks someone else will do it. On Thursday, the theory part is still empty.

Option B

Two members both start

They write the same section twice, and nobody starts the comparison table.

Like the radar operator in 1940, Amy has the information. The hard part is getting it to the right person, in a form that person can act on.

Example from our own group work.

Lesson 1: Keep one shared tracker

Run a group project through the same four steps as Dowding's network

Step	In 1940	In our project
Report	*Radar and observers reported to one network*	Post every update in one shared tracker, not in side chats
Filter	*The Filter Room removed duplicate plots*	One coordinator merges duplicate or conflicting updates twice a week
Display	*The tote board used fixed words*	Use four status words: Not started, Drafting, Ready for review, Done
Order	*"Vector 230, Angels 20"*	Write each request as owner, task and deadline: "Ming: theory slides, Thursday 8 pm"

Example tracker

Task	Owner	Status	Due
Story	Amy	Done	Tue
Theory slides	Ming	Drafting	Thu
Comparison table	Jason	Not started	Thu

Everyone can see that the comparison table has an owner but has not started.

Evidence In a flight simulation study, pairs who shared an understanding of the task and of each other worked together better and performed better.

For students Rotate the coordinator each week, so one person does not do all the filtering.

Mathieu et al. (2000); Dowding (1946)

Lesson 2: Confirm what you heard

Dowding's network checked its information with feedback, which Shannon and Weaver's model leaves out

1. **Listen** — Read the whole request.
2. **Paraphrase** — Repeat it in your own words.
3. **Check** — Ask: “Did I get that right?”
4. **Act** — Start once the other person confirms.

Amy's chat, with a paraphrase

Amy: *Can you do the theory part?*

Ming: *So you need the two theory slides by Thursday, with APA citations?*

Amy: *Yes, and the comparison table too.*

Evidence In a study of 115 people, speakers felt more understood when the listener paraphrased and asked follow-up questions than when the listener gave advice or only short replies.

For students Use it for every request with a deadline. A paraphrase checks the request without saying the other person was unclear.

Adapted from Rogers & Farson (1957/2021); Weger et al. (2014)

Tomorrow's Leaders

Germany had the aircraft. Britain had the network.

The network got the right information to the right person in time, which is our definition of effective communication.

- Send every update to one shared tracker
- Write each request as owner, task and deadline
- Confirm what you heard before you act

Questions for you

Where does information get lost in your group projects?

Who in your team merges the updates, the way the Filter Room did?

References

Battle of Britain Bunker. (n.d.). *Virtual bunker*. Retrieved November 11, 2026, from https://battleofbritainbunker.co.uk/virtual-resources/

The Battle of Britain: A not so narrow margin. (2015). *Air Power Review, 18*(2). https://www.raf.mod.uk/what-we-do/centre-for-air-and-space-power-studies/aspr/apr-vol18-iss2-7-pdf/

Boog, H. (1991). The Luftwaffe and the Battle of Britain. In H. Probert & S. Cox (Eds.), *The battle re-thought: A symposium on the Battle of Britain* (pp. 18–32). Airlife. https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Bracknell-No-1-Battle-of-Britain.pdf

Bournemouth University. (n.d.). *The Daventry demonstration*. Retrieved November 11, 2026, from https://histru.bournemouth.ac.uk/Oral_History/Talking_About_Technology/radar_research/the_daventry_demonstration.html

Bridge, P/O. (n.d.). *The Royal Observer Corps, 1939-1945* [Photograph]. Imperial War Museums (CH 8215). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:The_Royal_Observer_Corps,_1939-1945._CH8215.jpg

Cardon, P. W. (2008). A critique of Hall's contexting model: A meta-analysis of literature on intercultural business and technical communication. *Journal of Business and Technical Communication, 22*(4), 399–428. https://doi.org/10.1177/1050651908320361

Craig, R. T. (1999). Communication theory as a field. *Communication Theory, 9*(2), 119–161. https://doi.org/10.1111/j.1468-2885.1999.tb00355.x

Danielstirland. (2010). *Plotting table* [Photograph]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Plotting_Table.jpg

Daventry, B. J. (1940). *Supermarine Spitfire Mark IAs of No 610 Squadron, Royal Air Force based at Biggin Hill, Kent, flying in three 'vic' formations* [Photograph]. Imperial War Museums (CH 740). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Supermarine_Spitfire_Mark_Is_of_No._610_Squadron_based_at_Biggin_Hill,_flying_in_%27vic%27_formation,_24_July_1940._CH740.jpg

Devon, S. A. (1940). *Pilots of No 19 Squadron, Royal Air Force 'scramble' for the photographer, from the back of a lorry at Fowlmere* [Photograph]. Imperial War Museums (CH 1398). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Pilots_of_No._19_Squadron_RAF_stage_a_mock_%27scramble%27_from_the_back_of_a_truck_at_Fowlmere_near_Duxford,_September_1940._CH1398.jpg

References (continued)

Dowding, H. C. T. (1946, September 10). The Battle of Britain [Despatch]. *Supplement to The London Gazette*, (37719), 4543–4571. https://www.thegazette.co.uk/London/issue/37719/supplement/4543

Friends of the Battle of Britain Monument. (n.d.). *Key facts about the Battle of Britain*. Retrieved November 11, 2026, from https://bbm.org.uk/the-battle/battle-key-facts/

Gleave, T. (1991). August and September 1940. In H. Probert & S. Cox (Eds.), *The battle re-thought: A symposium on the Battle of Britain*. Airlife. https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Bracknell-No-1-Battle-of-Britain.pdf

Glines, C. V. (1990, September). Their finest hour. *Air Force Magazine*. https://www.airandspaceforces.com/article/0990finest/

Hall, E. T. (1976). *Beyond culture*. Anchor Press/Doubleday.

Historic England. (2020). *Chain Home transmitter tower, former RAF Stenigot* (List entry 1259301). https://historicengland.org.uk/listing/the-list/list-entry/1259301

Imperial War Museums. (2020). *Large print guide: Battle of Britain, Ops Block*. https://www.iwm.org.uk/sites/default/files/files/2020-09/OPS_BLOCK_Largeprint.pdf

Imperial War Museums. (n.d.-a). *Battle of Britain Big Wing: Was the Big Wing a bad idea?* Retrieved November 11, 2026, from https://www.iwm.org.uk/history/battle-of-britain-big-wing-was-the-big-wing-a-bad-idea

Imperial War Museums. (n.d.-b). *How the Observer Corps helped win the Battle of Britain*. Retrieved November 11, 2026, from https://www.iwm.org.uk/history/battle-of-britain-observer-corps

Imperial War Museums. (n.d.-c). *What was the Dowding System?* Retrieved November 11, 2026, from https://www.iwm.org.uk/history/what-was-the-dowding-system

References (continued)

Lunenburg, F. C. (2010). Communication: The process, barriers, and improving effectiveness. *Schooling, 1*(1), 1–10.

Mathieu, J. E., Heffner, T. S., Goodwin, G. F., Salas, E., & Cannon-Bowers, J. A. (2000). The influence of shared mental models on team process and performance. *Journal of Applied Psychology, 85*(2), 273–283. https://doi.org/10.1037/0021-9010.85.2.273

Ministry of Information. (n.d.). *Air Chief Marshal Sir Hugh C T Dowding, Commander in Chief of RAF Fighter Command during the Battle of Britain* [Photograph]. Imperial War Museums (D 1417). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Hugh_Dowding.jpg

Murphy, H. A., Hildebrandt, H. W., & Thomas, J. P. (1997). *Effective business communications* (7th ed.). Irwin/McGraw-Hill.

Price, A. (2003). Battle of Britain Day. *Royal Air Force Historical Society Journal*, (29). https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Journal-29-Dr-Alfred-Price-on-Battle-of-Britain-Day.pdf

Rogers, C. R., & Farson, R. E. (2021). *Active listening*. Mockingbird Press. (Original work published 1957)

Royal Air Force Museum. (n.d.-a). *Dowding: Architect of victory* [Podcast transcript]. Retrieved November 11, 2026, from https://www.rafmuseum.org.uk/documents/podcasts/transcripts/Dowding-Architect-Of-Victory.pdf

Royal Air Force Museum. (n.d.-b). *RADAR: The battle winner?* Retrieved November 11, 2026, from https://www.rafmuseum.org.uk/research/online-exhibitions/history-of-the-battle-of-britain/radar-the-battle-winner/

Royal Air Force Museum. (n.d.-c). *Sector room clock (replica)* [Museum object 71142]. Retrieved November 11, 2026, from https://collections.rafmuseum.org.uk/collection/object/object-71142/

Royal Air Force official photographer. (1940). *Flying Officer John Allen of No 54 Squadron receives the DFC from the King at Hornchurch* [Photograph]. Imperial War Museums (CH 442). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Royal_Air_Force_1939-1945-_Fighter_Command_CH442.jpg

References (continued)

Royal Air Force official photographer. (1945). *Chain Home radar installation at Poling, Sussex* [Photograph]. Imperial War Museums (CH 15173). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Chain_Home_radar_installation_at_Poling,_Sussex,_1945._CH15173.jpg

Royal Air Force official photographer. (n.d.). *The Operations Room at Headquarters, Royal Air Force Fighter Command, at Bentley Priory* [Photograph]. Imperial War Museums (C 1869). Wikimedia Commons. https://commons.wikimedia.org/wiki/File:The_Home_Front_in_Britain_during_the_Second_World_War_C1869.jpg

Shannon, C. E., & Weaver, W. (1949). *The mathematical theory of communication*. University of Illinois Press.

Spitzberg, B. H. (2013). (Re)Introducing communication competence to the health professions. *Journal of Public Health Research, 2*(3), Article e23. https://doi.org/10.4081/jphr.2013.e23

Weger, H., Jr., Castle Bell, G., Minei, E. M., & Robinson, M. C. (2014). The relative effectiveness of active listening in initial interactions. *International Journal of Listening, 28*(1), 13–31. https://doi.org/10.1080/10904018.2013.813234

Wood, D. (1991). The Dowding system. In H. Probert & S. Cox (Eds.), *The battle re-thought: A symposium on the Battle of Britain*. Airlife. https://www.rafmuseum.org.uk/documents/Research/RAF-Historical-Society-Journals/Bracknell-No-1-Battle-of-Britain.pdf